

Daniil Vlasenko

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🌐 daniilvlasenko.info

Research interests: cognitive and social sciences
Methodological interests: network science, data analysis, mathematical modeling, visualization of qualitative and quantitative information

Education

HSE University, Institute for Cognitive Neurosciences
PhD, Doctoral School of Cognitive Science
Moscow, Russia, November 2025 – present

HSE University, Institute for Cognitive Neurosciences
MS, Cognitive Sciences and Technologies
Moscow, Russia, September 2023 – June 2025

Saint Petersburg State University, School of Mathematics and Mechanics, Department of Statistical Modelling
BS, Applied Mathematics and Computer Science
St. Petersburg, Russia, September 2019 – June 2023

Additional Education

EITN EBRAINS Fall School in Computational Neuroscience. Attended with my wife – both independently selected.
Pavia, Italy, 21–28 September 2026

CNeuro, Theoretical and Computational Neuroscience Summer School.
Beijing, China, 7–14 July 2026

Neuromatch Academy, NeuroAI course.
Online, 14–25 July 2025

Neuromatch Academy, Computational Neuroscience course.
Online, 8–26 July 2024

Mediterranean School of Complex Networks.
Grado, Italy, 30 June – 5 July 2024

Bioinformatics Institute, Professional retraining program "Algorithmic Bioinformatics".
St. Petersburg, Russia, September 2022 – January 2023

Selected Publications

Vlasenko, D., Saranskaia, I., & Zakharov, D. (2026). Hypergraphs from multivariate connectivity: caCOH-based EEG/MEG representation. EPJ Special Topics.

Additional publications can be found on my Google Scholar page.

Selected Conferences

Bernstein Conference. Poster: “Hypergraphs from multivariate connectivity: caCOH-based EEG/MEG representation”. **Travel grant** recipient.
Frankfurt am Main, Germany, 29 September – 1 October 2026

NetSci 2025. Poster “Ensemble-Based Graph Representation of fMRI Data for Cognitive Brain State Classification”.
Maastricht, the Netherlands, 2–6 June 2025

Scholarships and Fellowships

Russian National PhD Research Scholarship, the program supporting PhD students and their dissertation research.
August 2026 – present

Combined Master's-PhD track at HSE University, the talent program designed for students enrolled on full-tuition scholarships.
September 2023 – present

Academic Development Program at HSE University, the talent management program aimed at supporting the professional development of promising researchers.
January 2024 – May 2025

Honors and Awards

Bernstein Conference **Travel Grant**
September 2026

Winner (1st Place), Information Technology & Mathematics Section, 10th All-Russian Youth Scientific Forum “Science of the Future – Science of the Youth”.
September 2025

Certificate of Excellence “**Best Diploma of the Year 2025**”, HSE University, Institute for Cognitive Neuroscience.
June 2025

Selected Research Experience

Agence Nationale de la Recherche project “DyndriteSymphony: Dynamics of nonlinear dendritic computation in cortical pyramidal neurons and interneurons”
External collaborator
September 2025 – present

My tasks: developing mathematical models of synaptic clustering that explain experimental observations, and investigating how passive dendrites can implement nonlinear computations.

Russian Science Foundation Interdisciplinary Grant 24-68-00030 "Next Generation Cognitive Artificial Intelligence"
Research-assistant
June 2024 – present

My tasks: developing graph and hypergraph representations of fMRI and EEG/MEG data, and applying network analysis and network-based machine learning to brain-state classification.

Teaching Experience

Supervision of course and diploma works of undergraduate students, **HSE University**
September 2024 – present

Seminar instructor (a few classes per course), “Advanced Computational Neuroscience”, “Some Topics of Calculus, Differential Equations and Nonlinear Dynamics”, **HSE University**
September 2024 – present

Languages

Russian, English

Skills

Mathematics | Probability and statistics; linear algebra; graph theory; optimization; calculus; (partial) differential equations; modeling.

Data analysis & machine learning | Analysis of quantitative and categorical variables; network analysis; machine (deep) learning; EEG/MEG, fMRI and neural electrode data analysis.

Programming & tools | Python: NumPy, Pandas, SciPy, scikit-learn, PyTorch (Geometric), MNE, Nilearn, NetworkX/iGraph; R: tidyverse, ggplot2; Git/GitHub; Linux/Bash; LaTeX.

Additional Interests

Bouldering (up to 6c/6c+), visual arts, non-fiction reading